Housing First or Treatment First?

A stylized model, and the quantitative model behind it

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1 A stylized model

1.1 Three questions

In the data, drug use is concentrated among low income households, most of the homeless do not carry a substance use disorder (SUD), and housing vouchers reach only about a quarter of the households eligible for them. The quantitative model is built to match facts of that kind. It has six state variables and fifty four parameters, which is what the welfare and fiscal questions require, but it also makes it hard to see which pieces of the environment do the heavy lifting. This note strips the environment down to help understand the main drivers in the model, but where my three main questions are still well posed.

Households are perfectly forward looking, here and in the quantitative model. They face the true hazard of acquiring a SUD, they know what carrying one would cost them, and they make no mistakes. Nothing below rests on misperception, on myopia, or on a taste that changes once the household has used.

1. **Who uses drugs?** In the data, use is concentrated at the bottom of the income distribution. Drugs here are a *luxury* good bought at a price: the Stone-Geary form in equation (1) below means the share of income spent on them rises with income, so left to itself the environment concentrates use at the *top*. Which feature of the environment reverses the concentration from the top to the bottom?
2. **How does housing assistance shape the drug use decision?** A voucher pulls drug use in two opposite directions. It raises the value of staying clean, because a housed future is worth more than a homeless one and using puts that future at risk: call this the value of life force, and it lowers use. In contrast, it cushions the consequence of acquiring a SUD, because a household that acquires one is housed anyway: call this the moral hazard force, and it raises use. Both are first order, so theory does not sign the effect of housing assistance on drug use, and the question is quantitative.
3. **When is Treatment First better than Housing First?** In a model without externalities, never. Housing First always raises welfare when agents are forward looking, as they are here. A SUD creates a genuine tension, since the same voucher that insures a household also makes the risk it insures more attractive, but that tension does not raise welfare, because Treatment First only removes an option that a household can take under Housing First as well. However, if a SUD and homelessness generate externalities, such as affecting the earning potential of the children of parents with a SUD, then Treatment First may be preferred: children of a parent with a SUD are themselves two to four times as likely to develop one (McGovern et al., 2023), and moving a child to a lower poverty neighbourhood before age thirteen raises adult earnings by 31 percent (Chetty et al., 2016). The model's work here is to show why Housing First raises welfare whenever there are no externalities, and then to qualitatively price the externality that overturns it.

The model is qualitative throughout: the calibration is rough by design, there is no validation exercise, and every component that obscures the mechanisms has been removed and the model refitted without it. What remains are the main drivers of the results. The welfare evaluations below are for illustration only; the full quantitative model is what settles the main question, which is whether Treatment First raises welfare relative to Housing First in a model that fits the data.

1.2 The environment

1.2.1 Households and preferences

There are two periods and a continuum of households. A period stands for thirty years of adult life, so the two of them span ages twenty five to eighty five. Parameters with a natural annual counterpart are

compounded to that length: the discount factor is β per period, the gross interest rate is $R = 1/\beta$ so that $\beta R = 1$, and ϕ_0 is survival over thirty years. Notation follows the quantitative model throughout.

A household is characterised by its productivity y, fixed over its life and lognormally distributed across households, $\log y \sim \mathcal{N}(-\frac{1}{2}\sigma_y^2, \sigma_y^2)$, so mean income is one. Households differ in y, in age $j \in \{1, 2\}$, in employment status $e \in \{0, 1\}$, in voucher receipt $\eta \in \{0, 1\}$ and in SUD status $d \in \{0, 1\}$, all discussed below, and in nothing else.

Flow utility depends on non durable consumption c, housing services h and drug consumption v,

$$u(c, h, v) \;=\; (1 - \gamma - \theta) \log c \;+\; \gamma \log h \;+\; \theta\big[\log(b + v) - \log b\big] \;+\; \aleph. \tag{1}$$

The three weights sum to one, so γ is the housing budget share. Drugs enter only in the first period, since that is the only period in which v is chosen, and housing is chosen each period from a three point grid $\{h_1, h_2, h_3\}$. The pair (θ, b) governs the drug term: the marginal utility of the vth unit is $\theta/(b + v)$, so θ mostly sets how much a user uses and b whether anyone uses at all. The Stone-Geary form keeps the first unit's marginal utility finite, which is what generates non users at all, because under plain $\theta \log v$ every household uses whatever the consequences. Finally $\aleph$ is the flow value of being alive, the utility of surviving into a period over and above what is consumed in it, which gives mortality a cost that does not vanish when consumption is low. Households save in one risk free asset and cannot borrow.

1.2.2 Timing

Period 1. Every household is clean. Employment $e \in \{0, 1\}$ is drawn with the unemployment probability u_0, and a household that is income eligible draws a voucher with probability π_V; these are the same two lotteries it faces again in period 2. It then chooses consumption, housing, drug use $v \geq 0$, which for simplicity is the only period in which it chooses whether to use, and saving, paying market rent p_r, which is normalised to one, on whatever unit it picks. It starts with no assets. Using drugs generates the risk of a SUD, increasing in v,

$$\pi(v) \;=\; 1 - e^{-\lambda v}. \tag{2}$$

Because π is increasing in v, the household internalises that using more is riskier, so the intensive and extensive margins are not separate decisions.

The transition. The SUD $d' \in \{0, 1\}$ is realised with probability $\pi(v)$ for households that used and with probability zero for the rest, and the household survives into period 2 with probability

$$\phi(d') \;=\; \phi_0\,(1 - \kappa d'), \tag{3}$$

so κ is the proportional reduction in survival caused by the SUD. The mortality penalty applies here, and here only.

Period 2. Terminal, after which the household dies with certainty. A household without the SUD is unemployed with probability u_0 and one with the SUD with the higher probability u_1; if unemployed it receives income support. The voucher is drawn again, subject to the policy rule. The household chooses consumption and housing, there is no drug use decision, and assets are run down with nothing bequeathed.

Mortality, and where the SUD is realised. Mortality is a tax on a continuation value, so the SUD can only cost anything if a future remains when it is carried. That does not require a third period. It requires the SUD to be drawn at a *transition* rather than inside the terminal period, and here it is drawn on the way into period 2, with survival over that transition depending on it, so acquiring it puts the whole of period 2 at risk. Realise it inside the last period instead and death costs nothing because nothing follows it, and the mechanism the model rests on falls silent.

1.2.3 Employment, income support and housing

The gap $u_1 - u_0$ is the entire resource cost of the SUD. Income in a period, given permanent productivity y and employment status e, is

$$y^g(y,e) \;=\; \begin{cases} y & \text{if } e=1, \\ \psi \cdot \min\{y, \bar{y}\} & \text{if } e=0, \end{cases} \tag{4}$$

with ψ the replacement rate and $\bar{y}$ a cap on the earnings base. There is no other income risk. The level of ψ is what links the SUD to homelessness: for the poorest households half of income can fall short of the rent on the smallest unit, so an unemployed household there cannot house itself out of current income.

Each period the household chooses

$$h_k \in \{h_1,\ h_2,\ h_3\}, \qquad h_1 < h_2 < h_3, \tag{5}$$

homelessness (h_1), a small unit (h_2) and a standard unit (h_3). Homelessness is free and delivers less housing services than the market does.[1]

A voucher covers part of the rent, is means tested and is rationed. It is drawn independently each period, which is a simplification relative to the quantitative model, where a voucher persists as long as eligibility holds. Writing $\eta \in \{0,1\}$ for whether one is held,

$$\Pr(\eta' = 1 \mid y', d', v) \;=\; \pi_V \cdot \mathbf{1}\{y' \le y_V\} \cdot \mathbf{1}\{\text{policy covers } (d', v)\}, \tag{6}$$

and a household holding one pays the fraction κ_r of market rent, so the rent paid on option k is

$$r_k(\eta) \;=\; p_r\, h_k \big[\kappa_r \eta + (1-\eta)\big] \cdot \mathbf{1}\{k \neq 0\}, \tag{7}$$

which is zero when the household is homeless. Vouchers operate in both periods, and π_V, the share of eligible households actually covered, is the policy dial of Section 1.5.

The two policies differ in exactly one respect, the last indicator in equation (6). **Housing First** covers every eligible household regardless of (d', v). **Treatment First** covers only households that satisfy the sobriety condition, $v = 0$ and $d' = 0$, which is the rule in the quantitative model as well. Since policy coverage in the states that follow drug use is the only difference, the gap between them is the moral hazard and the whole of it.

The condition binds in period 1 as well, and the timing there is worth stating. Every household arrives clean, so the period 1 voucher is drawn before v is chosen and conditionality cannot screen anyone out on entry. The order within the period is therefore: the household chooses v, the policy rule is then applied to the voucher it drew, and it chooses housing knowing what rent it faces. So under Treatment First a household that uses would be subject to market rent immediately and again in period 2.

1.2.4 The household's problem

Let a be assets carried in. Period 2 is terminal, so the household consumes everything and chooses housing,

$$W_2(a,y,e,\eta,d) = \max_k \Big\{ (1-\gamma-\theta)\log c + \gamma \log h_k + \aleph \Big\}, \qquad c = Ra + y^g(y,e) - r_k(\eta), \tag{8}$$

and write $W_2(a,y,d) \;=\; \mathbb{E}\big[W_2(a,y,e,\eta,d) \;\mid\; y, d\big]$ for its expectation over the period 2 employment and voucher draws. In period 1 the household draws (e,η) and then chooses housing, drug use and saving,

$$V_1(y,e,\eta) = \max_{k,\, v\ge 0,\, a'\ge 0} \Big\{ (1-\gamma-\theta)\log c + \gamma \log h_k + \theta[\log(b+v) - \log b] + \aleph + \beta\, \mathbb{E}_{d'} \Big[\phi(d')\, W_2(a', y, d') \Big] \Big\}, \tag{9}$$

with $c = y^g(y,e) - r_k(\eta) - p_v v - a'$ and $d' = 1$ with the use dependent probability of (2). Survival sits inside the expectation over d' because it depends on which d' is realised.

A household uses if the value at its best positive v exceeds the value at $v = 0$. Households differ only in y and in the draw (e,η), which is the same lottery for everyone, so within a draw the decision defines a set of income intervals on which use occurs, and participation at a given income is the probability of drawing a state in which the household uses.

[1] Two rented sizes is the fewest that lets a household economise on housing rather than lose it. With one, homelessness comes out at 14.40 percent against a target of 0.40, at any h_1.

1.3 Calibration

1.3.1 External calibration

Table 1 lists the parameters set outside the model. Preferences and demographics are conventional: the discount factor over a thirty year period is $\beta = 0.860$[2] and the gross interest rate is $R = 1/\beta$, the dispersion of log permanent income is $\sigma_y = 0.75$ from Heathcote, Perri and Violante (2010), and survival over a period is $\phi_0 = 0.91$, from the Social Security Administration (SSA) period life table for twenty five to fifty five. For the SUD, the proportional fall in survival is $\kappa = 0.49$, derived by compounding an annual mortality ratio of 8, the midpoint of the range in Hjorthøj et al. (2023), so a clean household's chance of dying before period 2 is 9 percent and a household that acquires a SUD faces 53 percent, and the two unemployment probabilities are measured directly. $u_0 = 0.049$ is the headline civilian unemployment rate from the Bureau of Labor Statistics (BLS) Current Population Survey, series UNRATE as distributed by Federal Reserve Economic Data (FRED), averaged over 2014 to 2024. And $u_1 = 0.1625$ is from the National Survey on Drug Use and Health (NSDUH) 2024 for respondents aged 26 and over.

Income support replaces half of previous earnings on a base capped at mean income, in line with statutory state rules. The two rental sizes are set such that the standard unit costs 0.41 of mean income and the small unit is a quarter of it. The voucher parameters are institutional: covered tenants pay 30 percent of market rent, eligibility is capped at mean income, and $\pi_V = 0.27$ is the assisted eligible share in the 2023 Picture of Subsidized Households, published by the Department of Housing and Urban Development (HUD). The weight on drugs is set at $\theta = 0.23$.[3]

1.3.2 Internal calibration

Five parameters are set inside the model, in Table 2, each bracketed on the moment it moves most rather than fitted jointly. The weight on housing γ is matched to the 26.8 percent of total expenditure that shelter plus utilities take in the 2024 Consumer Expenditure Survey (CEX), housing over expenditure net of drugs. The flow value of being alive $\aleph$ is matched to a value of a statistical life of 80 times average annual income, about \$9 million against a mean household income of \$113,000, which is the range surveyed by Viscusi and Aldy (2003) once their estimates are put in current dollars. The pull of the first unit b is matched to a drug participation rate of 5.13 percent from NSDUH 2024 for respondents aged 26 and over, housing services on the street h_1 to a homelessness rate of 0.399 percent from the 2023 HUD point in time (PIT) count, converted to households, and the SUD hazard λ to the 33.2 percent of users who acquire a SUD.

1.4 Who uses drugs

This section asks which feature of the model puts drug use at the bottom of the income distribution, and it starts from the force that any answer has to overcome. Drugs are a luxury good bought at a price, so a household with twice the income gives up less utility per dollar spent on them. With no consequences at all for acquiring a SUD that is the only force operating, and drug use concentrates at the top: the average user then earns 2.62 times the population average.

Two features of the environment are candidates to reverse it. The *mortality channel* is that the SUD increases the chance of dying, through κ in equation (3). The *unemployment channel* is that it costs you a job with probability $u_1 - u_0$, and if income support cannot cover the rent you downsize, live on less, or lose your home. Figure 1 scales each from nothing to 125 percent of its calibrated value, recalibrating b at every point so participation stays at 5.13 percent. The amount of drug use is therefore identical all along both curves and the only thing moving is which households use.

[2]The annual discount rate here is higher than the one in the quantitative model, and this is deliberate. A period here is thirty years, so every consequence of using drugs arrives thirty years after the decision, whereas in the quantitative model consequences arrive annually and compound. A low annual discount factor would weigh that distant death much less and the mortality channel would be losing to impatience rather than to anything the model is about, so a higher factor corrects for the coarseness of the horizon rather than making a claim about preferences.

[3]A θ large enough to match the eleven percent of income that users spend on drugs makes the drug too valuable to be deterred by a death that arrives thirty years later, and users stop being poor. Setting $\theta = 0.23$ gives a spending share of 6.32 percent and keeps the income gradient, which is the trade this model can make.

Table 1: Externally set parameters

Parameter	Description	Source or rationale	Value
Preferences and demographics			
β	Discount factor over a thirty year period	0.99 annually, compounded	0.860
R	Gross interest rate over a thirty year period	Set to $1/\beta$	1.162
σ_y	Dispersion of log permanent income	Heathcote et al. (2010)	0.75
ϕ_0	Survival over a period, no SUD	SSA period life table, ages 25–55	0.91
θ	Weight on drug use	Set; see the text	0.23
The SUD			
κ	Fall in survival with a SUD, over a thirty year period	Hjorthøj et al. (2023), age 26+	0.49
u_0	Unemployment probability, no SUD	BLS CPS via FRED, 2014 to 2024	0.0485
u_1	Unemployment probability, with a SUD	NSDUH 2024, age 26+	0.1625
Income support			
ψ	Replacement rate while unemployed	DOL, statutory UI rates	0.50
$\bar{y}$	Cap on the earnings base, multiple of average income	DOL, state benefit maxima	1.00
Housing, drugs and the voucher			
p_r	Rent per unit of housing services	Normalisation	1.00
p_v	Price of a unit of drugs	Normalisation	1.00
h_3	Standard rental unit	Set	0.41
h_2	Small rental unit	Set	$0.25 \times h_3$
κ_r	Tenant's share of the rent when covered	HUD voucher rules	0.30
y_V	Voucher means test, multiple of average income	HUD income limits	1.00
π_V	Voucher generosity in the baseline	HUD 2023, assisted eligible share	0.27

Table 2: Internally calibrated parameters

Parameter	Description	Target moment	Data	Model	Value
γ	Weight on housing services	Housing over non drug spending	26.80%	26.58%	0.289
$\aleph$	Flow value of being alive	Value of a statistical life	80.0	80.0	2.790
b	Pull of the first unit of drugs	Drug participation rate	5.13%	5.13%	0.00316
h_1	Housing services when homeless	Homelessness rate	0.40%	0.42%	0.00573
λ	Addiction hazard per unit of use	Share of users who acquire a SUD	33.2%	35.5%	132.6

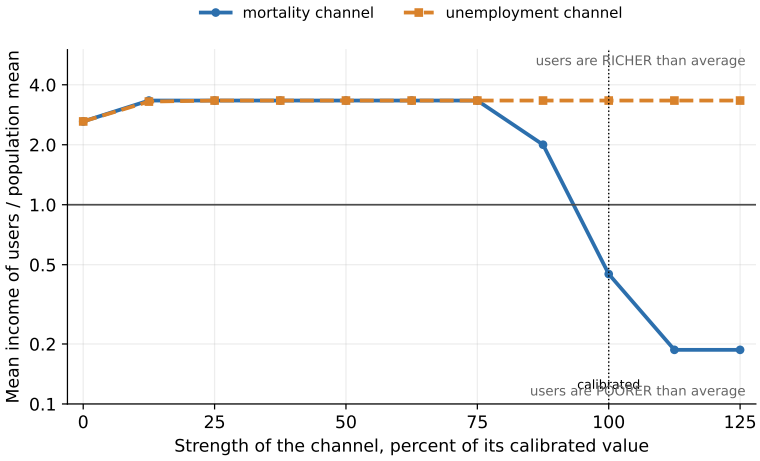


Figure 1: What it takes to reverse the income gradient in drug use

Note: the vertical axis is average income of users over the population average, on a log scale, and b is recalibrated at every point so that participation is 5.13 percent throughout. The black dotted line marks the calibrated strength; everything to the right of it is extrapolation.

The result is sharper than the statement that mortality matters more. That channel takes the income ratio down to 0.45 at full strength, and it does so as a threshold rather than gradually: at half of its calibrated

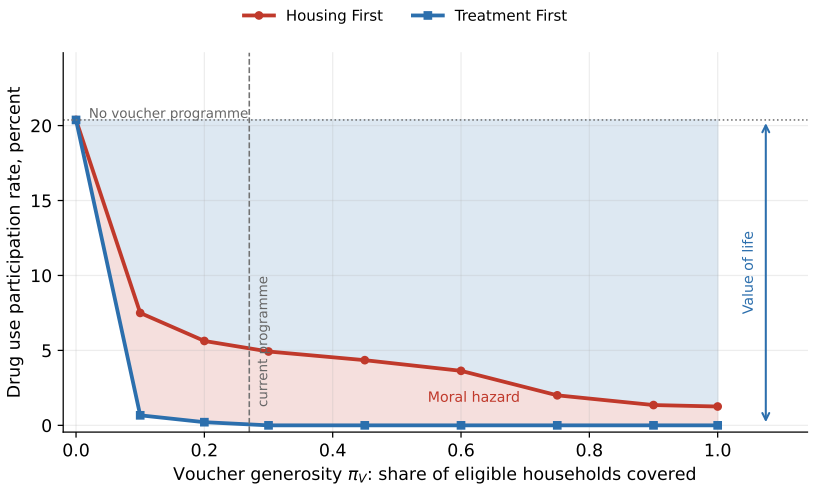


Figure 2: Drug use under the two policies as the voucher becomes more generous

Note: the red band is the moral hazard force and the blue band above it is what Housing First delivers net; together they are the value of life force, marked at the right. The dashed line is the generosity of the current programme, $\pi_V = 0.27$.

value the ratio is still high, at 3.33, and the reversal happens over the last tenth or so of the channel. Past the calibrated point it keeps biting, reaching 0.19 at 125 percent, so the reversal is not a knife edge that the calibration happens to sit on. The unemployment channel does not reverse it at all, leaving users at 3.33 times average income even fully switched on, and at 125 percent it is 3.33, still above one and no closer to turning. Strengthening the resource penalty by a quarter does not begin to do what mortality does. A resource penalty does not produce the concentration of drug use at the bottom.

A mortality penalty is different in kind. A resource penalty scales with the very thing it is meant to offset, since taking a fixed share of a large income still leaves a large income, so it cannot outrun the fact that drugs are cheaper in utility terms for the rich. What mortality destroys is not resources but a continuation value, and the continuation value rises with permanent income. The household with more to lose loses more, and loses proportionally more than it gains from drugs being relatively cheap. That is the work of $\aleph$ in equation (1): it is the part of the continuation value that does not shrink as consumption falls.

1.5 How does housing assistance shape the drug use decision?

The two forces of question 2 can be separated exactly, because they differ in what happens to a household that uses. Under Treatment First the voucher is withdrawn from any household that violates the sobriety condition, so only the value of life force operates. Under Housing First it is not, so both channels operate. Comparing the two policies at the same generosity therefore isolates the moral hazard force, and comparing Treatment First with no programme at all isolates the value of life force.

Figure 2 traces both as generosity rises. With no programme at all, 20.38 percent of households use. Universal Treatment First takes that to 0.00 percent, a value of life force of -20.38 percentage points; universal Housing First leaves it at 1.26 percent, so the moral hazard force is +1.26 points and the net effect of Housing First is -19.12. Both forces are first order and they work against each other, which is the central feature of the problem. Anyone asserting that housing the homeless must increase drug use, or must reduce it, is making a quantitative claim that needs to be tested.

Figure 3 plots the homelessness and SUD outcomes for each programme as its generosity changes, alongside drug use. In panel (a) the two policies end in the same place on homelessness, both driving it from 0.51 percent with no programme to zero, and along the way Treatment First is if anything slightly below Housing First, because the households it deters from using never acquire the SUD that puts their housing at risk. Panel (b) repeats Figure 2 here for convenience. The larger difference is in panel (c), where at universal cover Housing First leaves 0.217 percent of the population carrying the SUD and Treatment First

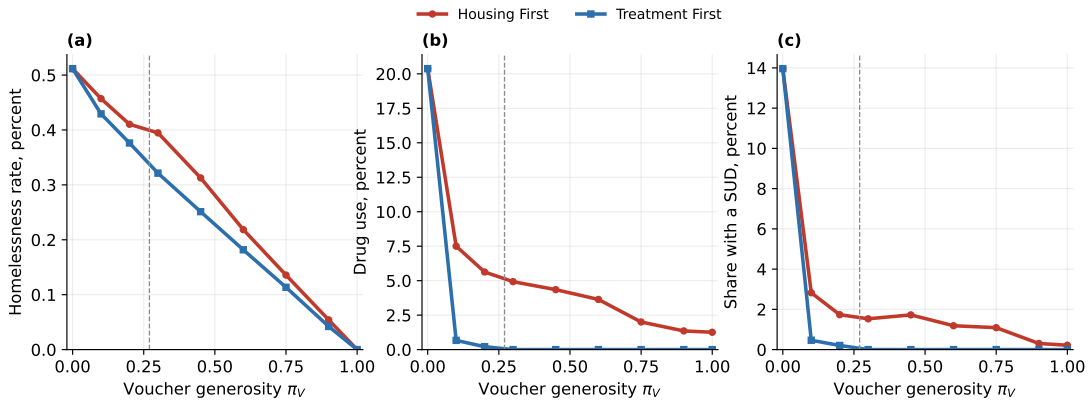


Figure 3: Homelessness, drug use and the SUD under the two policies

leaves 0.000 percent. What the moral hazard buys is extra SUD cases among households that are housed either way.

1.6 Welfare

1.6.1 Without externalities, Housing First always wins

Sections 1.4 and 1.5 establish what each policy does to behaviour. This section asks what each does to welfare, comparing the two at the same generosity π_V, whose outcomes are in Panel A of Table 3. The policies differ only in the last indicator in equation (6), so Housing First transfers resources in a superset of the states in which Treatment First does. The budget set is weakly larger in every state and strictly larger in one, so a household that optimises cannot be worse off under Housing First:

$$V_1(y;\ \mathrm{HF},\ \pi_V)\ \geq\ V_1(y;\ \mathrm{TF},\ \pi_V) \qquad \text{for every } y, \tag{10}$$

with equality exactly for households that never use. Put the other way round, a household under Housing First can reproduce its Treatment First outcome exactly by not using: conditionality removes an option from a forward looking optimiser and gives nothing back. This is a dominance argument, not a calibrated result, and no choice of parameters overturns it.[4]

It follows that the moral hazard is not a distortion. Households here face the true SUD hazard and make no mistakes, so the extra use Housing First induces is not a wedge to be closed but a household optimally taking a risk it is now better able to bear. The +1.26 points of Section 1.5 are a statement about drug use and not about welfare. Panel B of Table 3 puts numbers on it: 2.7 percent of households are strictly worse off under Treatment First, none is better off, and those affected lose 0.50 percent of consumption. Those are exactly the households that use: a household that never uses faces the same voucher rule under both policies and is indifferent between them. The figure is not the 5.13 percent participation rate of Section 1.3 because the programmes compared here are universal, rather than the rationed benchmark that rate is measured at.

A fixed budget does not rescue the welfare case for conditionality either. Treatment First is almost exactly as expensive as Housing First at universal cover, 0.3185 against 0.3184, so matching on cost rather than on generosity changes nothing. Two things offset. Survival into period 2 falls from 0.91 to 0.47 with a SUD, so the extra households Housing First puts into the SUD state die sooner and fewer are alive in period 2 to collect a voucher; against that, Treatment First withdraws the voucher from anyone who used, so it pays for fewer tenants while they live. The expected outlay remains the same. So a case for conditionality has to come from an externality.

Table 3: Welfare at universal cover, with no externality

	Housing First	Treatment First
Panel A. Outcomes at universal cover		
Drug participation rate	1.26%	0.00%
Share with a SUD	0.196%	0.000%
Homelessness rate	0.000%	0.000%
Voucher outlay per household	0.3185	0.3185
Panel B. Treatment First against Housing First, no externality		
Households strictly worse off		2.7%
Households strictly better off		0.0%
Consumption equivalent loss, those affected		0.45%
Consumption equivalent loss, whole population		0.012%

Note: the consumption equivalent is the proportional change in consumption, in every period and state, worth as much as the difference in lifetime value. Voucher outlay is expected spending on a household over both periods, the second discounted and weighted by survival, per unit of average income.

Table 4: The externality on children

Panel A. Transmission: what each income penalty implies			
δ_d	Child's SUD rate	Ratio	Loss to the child
0.00	2.24%	1.00	0.0%
0.20	4.04%	1.80	38.7%
0.30 (calibrated)	5.45%	2.43	69.2%
0.40	7.43%	3.31	113.3%
Panel B. Does the externality reverse the ranking?			
Private welfare loss from conditionality			0.012%
Disorder cases prevented			0.196 pp
Damage to the child of one such case			54%
Externality gain from conditionality			0.106%
Gain over loss			8.9

Note: Panel A is at the baseline programme, $\pi_V = 0.27$, since the transmission ratio in the data comes from an economy with a rationed programme; Panel B is at universal cover, one child per household, no discounting across generations. Losses are consumption equivalents.

1.6.2 An externality on children reverses it

I model an externality that runs through income rather than through preferences. I have considered two alternatives, but both seem to be at odds with reality. Letting the child of a user inherit a stronger taste, a lower b, matches the transmission fact but has the wrong welfare sign: b enters only through $\theta \log(1 + v/b)$, so a lower one raises the child's objective at every $v > 0$ and its lifetime value with it. So the specification says that a child who inherits a stronger taste for drugs is *made better off* by inheriting it, and that is the opposite of what these children report. Growing up with household substance abuse is one of the adverse childhood experiences in Felitti et al. (1998), and it predicts higher rates of depression, worse self reported health and lower life satisfaction in adulthood; the children of parents with a SUD also have lower educational attainment, which is the premise of the transmission literature δ_d is calibrated against (McGovern et al., 2023). A model in which the transmitted trait raises the child's own welfare cannot be describing any of that, whatever it does to the transmission rate. Letting it inherit a greater susceptibility, a higher λ, fails another way, since a child facing a riskier drug optimally uses *less* ex ante, which is at odds with the data: children of parents with a SUD are *more* likely to use, not less.

Income works on both counts, and it works through the channels explored in Section 1.4: lower permanent income is unambiguously bad to receive, and this model already puts users at the bottom, so a child born poorer is mechanically more likely to use, which is aligned with the data. It is also supported by the evidence, and it stands in for more than income alone: a parent who diverts spending and time to drugs invests less

[4]The same comparison in the quantitative model gives the same answer.

in the child's education and development, and the income penalty is the reduced form of all of that (Chetty, Hendren and Katz, 2016; McGovern et al., 2023). The income gradient does the whole job, so the model transmits a SUD across generations *without transmitting any preference for drugs.*

A child's baseline income is drawn from the same lognormal as everyone else, and the entire link is the penalty

$$y_{\text{child}} \;=\; y_{\text{base}}\,(1-\delta_d)^{\mathbf{1}\{\text{parent acquired the SUD}\}}\,(1-\delta_h)^{\mathbf{1}\{\text{parent was homeless}\}}, \tag{11}$$

which the parent does not internalise. That is what makes it an externality rather than something already priced into V_1.[5]

The two penalties are set to different things. Panel A of Table 4 shows that $\delta_d = 0.30$ makes the child of a parent who acquired the SUD 2.4 times as likely to acquire one, in the middle of the epidemiological range of two to four in McGovern et al. (2023) and the alcohol literature. The homelessness penalty δ_h is set to 31 percent, directly from Chetty et al. (2016), who find that children who moved to a low poverty neighbourhood before age 13 went on to earn 31 percent more as adults. The two penalties also do different work: both policies drive homelessness to zero at universal cover, so δ_h prices the programme against no programme while δ_d decides Treatment First against Housing First.

Panel B puts the two sides together and they are not close. The private cost of conditionality, in the first row, is 0.013 percent of consumption, small because only 2.7 percent of households are affected. Against that, conditionality prevents 0.217 percentage points of SUD cases in the second row, and each of those cases costs one child 54.3 percent of consumption, shown in the third row. The gain in the fourth row is the product of those two, since one prevented case spares one child: 0.217 percentage points times 54.3 percent gives 0.118 percent of consumption, which is 8.9 times the private loss of the first row. That ratio exceeds one for every δ_d in Panel A except $\delta_d = 0$, where by construction there is no externality to price. Housing First is better for every household taken one at a time, and worse once one counts the children of the households it fails to deter.

All of this is a crude illustration: the model is two periods long and nobody who faces the sobriety condition fails to meet it, so the impacts of policies can be unreasonably large. It shows that an externality can reverse a welfare ranking that is otherwise airtight, which is all it is for. Which way the comparison actually goes is a question about magnitudes, and that is what the quantitative model of the next section is for.

The quantitative model also carries more externalities than are described here, and they do not all pull the same way, which is another substantive reason the welfare ranking is a question of magnitudes rather than of signs.

- **Programmes are paid for out of distortionary taxes, and they shrink the base they are paid from.** A programme that houses more households costs more, raising taxes that distort the hours and the saving of every household, including those it never reaches. It also lowers the labour supply of the households it does reach, because eligibility is means tested: the voucher is clawed back at a rate of κ_r, so a recipient who works more loses some of the transfer. This means that other households have to pay more taxes, which further distorts their labour supply. Neither effect exists here, where hours are fixed and financing is not modelled.
- **Homelessness is a disamenity in itself.** Households dislike homelessness around them and not only their own, so the aggregate rate is an argument of everyone's utility. It runs in favour of whichever policy houses more, which is the opposite direction from the tax.

2 The quantitative model

This section describes the model the paper's numbers come from. Where the stylized model already carries something unchanged it is named rather than motivated a second time: the Stone-Geary form of the drug term, the logic that separates the two policies, and the transmission externality all appear below. Notation

[5] In the quantitative model the altruism of parents towards their children is calibrated rather than set to zero, so that the channel where a parent does take some account of what it does to its child is present and only the remainder is external.

follows Section 1 wherever the two models share an object: γ and θ are the housing and drug weights in both, b is the same Stone-Geary endowment, $\aleph$ the same flow value of being alive, λ the same SUD hazard, ψ and $\bar{y}$ the same replacement rate and earnings base cap, r the same risk free rate, and β is the discount factor and nothing else.

2.1 Demographics, the state and the choice set

Households enter as adults, live at most J annual periods and retire at J_R, and j indexes age within that span. Survival from j to $j+1$ is $\phi(j, h', d, m)$ in (24), and it depends on age, on the housing *chosen* this period, on the SUD and on serious mental illness (SMI). Newborns replace the dead exactly, so the mass of households is constant. The individual state is

$$s = (a,\, y,\, e,\, d,\, m,\, \eta) \in \mathcal{A} \times \mathcal{Y} \times \{1,2,3\} \times \{0,1\}^3, \tag{12}$$

where a denotes assets, y permanent productivity, e employment status, d the SUD, m SMI and η receipt of a voucher.

Each period the household chooses

$$\big(n,\, h',\, v,\, \tau,\, a'\big) \in [0,1] \times \{h_1, h_2, h_3, h_4\} \times \mathbb{R}_+ \times \{0,1\} \times \mathcal{A}, \tag{13}$$

where n denotes hours, h' the housing unit chosen, of which h_1 is homelessness, v drug use, τ SUD treatment take up, available only when $d = 1$ and requiring $v = 0$ in the period it is taken, and a' next period assets.

2.2 Preferences

Period utility is

$$u\big(c, h', v, n, \tau; d, m\big) \;=\; \frac{\big[\, c^{1-\gamma-\theta}\,(h')^{\gamma}\,(v+b)^{\theta}\,\big]^{1-\sigma}}{1-\sigma} \;+\; \omega_\ell \frac{(1-n)^{1-\sigma_\ell}}{1-\sigma_\ell} \;+\; \aleph \;-\; \tau\bar{\tau}^{u} \;-\; \kappa_x H, \tag{14}$$

Four of these are the stylized model's and carry the same meaning here as in Section 1.2: γ the housing weight, θ the drug weight, so that $1-\gamma-\theta$ is the weight on consumption and the three sum to one; b the Stone-Geary endowment; and $\aleph$ the flow value of being alive. Equation (1) is the logarithmic case of this bundle with hours fixed.

The rest are new, and there are three of them. Leisure makes hours a margin, at weight ω_ℓ and curvature σ_ℓ over the $1-n$ hours not worked. SUD treatment costs utility, $\bar{\tau}^u$ in each period the household takes it up, $\tau = 1$. And homelessness carries a disamenity, κ_x per unit of the aggregate homelessness rate H, borne by every household and priced on the headcount alone, which is what makes homelessness an externality here rather than only a private misfortune; Section 3.1 takes it up. Outside the bundle, σ is the coefficient of relative risk aversion.

2.3 Income, taxes and transfers

Gross income depends on age, employment, productivity and health,

$$y^{g}(j, y, e, d, m, n) = \begin{cases} 0.40\, y & j \geq J_R \quad \text{(social security)} \\ n\, y\, \varepsilon(j)\, \rho_d^{\mathbf{1}\{d=1\}} \rho_m^{\mathbf{1}\{m=1\}} & j < J_R,\ e = 1 \quad \text{(employed)} \\ \psi\, \min\{y\, \varepsilon(j),\ \bar{y}\} & j < J_R,\ e = 2 \quad \text{(first year of a spell)} \\ \zeta_{DI}\, y\, \varepsilon(j) & j < J_R,\ e = 3,\ m = 1 \quad \text{(non employed with SMI: disability)} \\ 0 & \text{otherwise,} \end{cases} \tag{15}$$

taking the cases in the order they appear. Retirement income replaces 40 percent of the productivity state with no health penalty, since social security is not means tested on health. For the employed, $\varepsilon(j)$ is

the deterministic age profile of efficiency and ρ_d and ρ_m are productivity penalties applied *conditional on working*. The three employment states are work, $e = 1$, the first period of a spell out of work, $e = 2$, and non employment thereafter, $e = 3$, so the last two cases are both unemployment and the disability case is the income of a household that is out of work and carries SMI. Unemployment insurance (UI) is paid in the first period of a spell only, at the replacement rate ψ on an earnings base capped at average income $\bar{y}$, both as in equation (4). ζ_{DI} is the rate at which disability insurance (DI) replaces the income of a non employed household with SMI.

The two penalties, ρ_d and ρ_m, are in the model to generate a large share of homeless households being employed, consistent with the data. Employed homelessness cannot be produced by unemployment alone: something has to leave a household poor while it is still in a job. Both are calibrated internally: ρ_d on the share of the homeless who are employed, and ρ_m on the earnings of workers with SMI relative to average.

Labour income is taxed with the Heathcote et al. (2017) function and capital income at a flat rate τ_k,

$$y^n = \tau_\ell \, (y^g)^{\xi} \; + \; (1 - \tau_k) \, r \, a, \tag{16}$$

with y^n denoting after tax income, where ξ governs progressivity and τ_ℓ the level. Households can receive two transfers, both means tested on gross income as the programmes themselves are. A food stamp floor tops means tested income up to $\underline{c}\,\bar{y}$, with $\underline{c}$ the floor expressed as a share of average income, and the voucher covers rent above a tenant contribution of κ_r of gross earnings, the same κ_r as in Section 1.2.3,

$$F = \max\big\{0, \; \underline{c}\,\bar{y} - (y^g + ra)\big\}, \qquad Vch = \mathbf{1}\{\eta = 1\}\,\mathbf{1}\{h' \neq h_1\} \, \max\big\{0, \; p_r \, h' - \kappa_r \, y^g\big\}. \tag{17}$$

A homeless voucher holder does not lose the voucher: retention depends on income eligibility, not on housing, so the entitlement survives and simply pays nothing until the household rents again.

2.4 The budget constraint

$$c + p_v v + a' \; = \; y^n + a \; - \; p_r \, h' \, \mathbf{1}\{h' \neq h_1\} \; + \; Vch + F, \qquad c > 0, \; v \geq 0, \; a' \geq 0. \tag{18}$$

The drug price is p_v and market rent is p_r, both as in Section 1.2.3, and a is assets carried in. Note what is absent: treatment, which costs utility, and bequests, which go to the government. How c, v and n are chosen out of it is with the rest of the optimisation, in Section 2.11.

2.5 Housing, and the homelessness loop

Housing is a four point grid. h_1 is literal homelessness, sleeping rough or in an emergency shelter, which is the HUD point-in-time definition and excludes doubling up; it carries a small service flow and no rent. The three rental sizes are geometric,

$$h_2 = \underline{h}, \qquad h_3 = \underline{h}\,\varsigma, \qquad h_4 = \underline{h}\,\varsigma^2, \tag{19}$$

so two parameters describe the grid: $\underline{h}$ the smallest rental size and $\varsigma > 1$ the geometric spread between consecutive sizes.

Homelessness then enters three separate equations: survival (24), the employment transitions (26) and the SUD recovery hazard (21), through $\kappa_{\phi H}(j)$ and κ_{ueH} in Section 2.8 and μ_H in Section 2.6. It raises mortality, worsens both employment margins and lowers the chance of recovering from the SUD. Mental illness is the one thing it does not reach. The SMI chain runs on its own, and that is a finding rather than an omission: the trial that housed people with serious mental illness looked for an effect on their symptoms and did not find one, while finding large effects on their housing. Section 2.7 gives it.

2.6 Drug use, the SUD, recovery and treatment

A household either abstains or uses, with v chosen continuously. A clean household that uses acquires the SUD with the hazard of equation (2),

$$\Pr(d' = 1 \mid d = 0, v) = \pi_d(v) = 1 - e^{-\lambda v}, \tag{20}$$

In the baseline a clean household may initiate only up to an age J_E, the last age at which use can begin.[6] A household with the SUD recovers with probability

$$\pi_r(h', m, \tau) = \min\left\{1,\ \bar{\pi}_r\, \mu_H^{\mathbf{1}\{h'=h_1\}}\, \mu_m^{\mathbf{1}\{m=1\}}\, g^{\tau}\right\}, \tag{21}$$

with $\bar{\pi}_r$ the baseline annual chance of recovering while housed, clean of SMI and untreated, and three multipliers on it: $\mu_H < 1$, so recovery is harder on the street, $\mu_m < 1$, harder with SMI, and $g > 1$ the treatment effect.

μ_H is how housing acts on drug use here, and it is worth being exact about what that claim is and is not. Housing does not change what a household wants: flow utility is the same function of (c, h, v) housed or not. What housing changes is the chance of getting out of the disorder once it is carried, recovery on the street being μ_H times recovery while housed. Milby et al. (2005) randomised cocaine dependent homeless adults into arms differing in housing and measured weeks of abstinence, so the ratio between arms identifies μ_H in our calibration.

Treatment τ is a *choice*, not a mechanical hazard, taken at the utility cost $\bar{\tau}^u$ in (14). It also costs the drug: a household in treatment must abstain in that period, $v = 0$ whenever $\tau = 1$. And g is the parameter through which its entire effect runs, so treatment has to be something a household decides to undertake and can decline.

2.7 Serious mental illness

m is an exogenous two state Markov chain representing SMI, whose prevalence p_m is about 5.5 percent of adults, defined by substantial functional impairment, rather than any mental illness, which is roughly a quarter of adults and includes mild anxiety: a state that halves productivity cannot sensibly cover a quarter of the population.

The onset rate is a function of age. Writing $p(j)$ for prevalence at age j and λ_m for the annual chance of leaving the state,

$$\text{onset}_m(j) = \frac{p(j{+}1) - p(j)(1 - \lambda_m)}{1 - p(j)}, \tag{22}$$

which is the accounting identity that reproduces the observed age profile exactly, backed out of the age profile of prevalence in NSDUH 2023 and the recovery hazard. The transitions are then

$$\Pr(m' = 1 \mid m = 0, d) = \text{onset}_m(j) \cdot \iota^{\mathbf{1}\{d=1\}}, \qquad \Pr(m' = 0 \mid m = 1) = \lambda_m, \tag{23}$$

where $\iota > 1$ is the multiplier on onset for a household carrying the SUD, which makes a SUD raise the hazard of SMI. That last channel runs in the direction the evidence supports: substance induced psychosis converts to schizophrenia or bipolar disorder at a pooled rate near a quarter (Murrie et al., 2020), and in the Danish registers any SUD predicts conversion to schizophrenia with a hazard ratio of 1.34, opioid use disorder 2.74 (Nielsen et al., 2017).

SMI then affects five objects: productivity conditional on working through ρ_m in (15), both employment margins through κ_{uem} in (26), mortality through $\kappa_{\phi m}$ in (24), the SUD recovery hazard through μ_m in (21), and disability income, for which eligibility is being out of work with SMI and the replacement rate is ζ_{DI}. It does not change the utility function.

The mental health chain does *not* depend on housing, and that is a finding in the data rather than an omission. In the At Home / Chez Soi randomised trial, 1,198 homeless adults with SMI across four cities, assignment to housing with intensive case management produced no detectable effect on symptom severity at any wave, while producing large effects on housing stability, its primary outcome. Making the channel housing dependent would mean adding a mechanism the best available trial looked for and did not find.

[6] Without it there is little consequence at old ages to using: too little life remains for the disorder to damage, so the model produces too much of it.

2.8 What the SUD, SMI, and homelessness cost

Death risk. Survival is

$$\phi(j,h',d,m) = 1 - \min\Big\{1,\ (1-\bar{\phi}(j))\,\kappa_{\phi H}(j)^{\mathbf{1}\{h'=h_1\}}\,\kappa_{\phi d}(j)^{\mathbf{1}\{d=1\}}\,\kappa_{\phi m}^{\mathbf{1}\{m=1\}}\Big\}, \tag{24}$$

with $\bar{\phi}(j)$ the baseline survival age profile and $\kappa_{\phi H}(j)$ and $\kappa_{\phi d}(j)$ two age varying multipliers, and $\kappa_{\phi m}$ one that is not, raising the death hazard for homelessness, the SUD and SMI respectively. Because $\aleph > 0$ and guarantees that the value of life is positive, a higher death hazard is a real cost, and it is larger for households with more life remaining and more to consume, which is precisely the sorting Section 1.4 isolates.

$\bar{\phi}(j)$ is the SSA period life table for ages 26 to 85, with the sexes combined by survivor weight, $q = [l^M q^M + l^F q^F]/[l^M + l^F]$.

The homelessness multiplier cannot be read off the data directly, and the reason is worth spelling out. Baggett et al. report a *standardised mortality ratio* (SMR) for homeless adults: deaths among the homeless at a given age divided by deaths in the general population at that age, which they publish by age band and which we write $RR(j)$. It cannot be used as $\kappa_{\phi H}(j)$ because the people it is measured on are not only homeless. A share $s = 0.217$ of them also carry a SUD (Gutwinski et al., 2021), which raises mortality by itself, so the published ratio is a mixture of two effects and attributing all of it to housing would overstate what housing does.

Three things are observed and two are solved for. Observed: $RR(j)$, Baggett's ratio; $\sigma_d(j)$, the shape of the SUD's mortality effect over age; and s, the SUD share of the homeless. Solved for: $\kappa_{\phi d}(j)$ and $\kappa_{\phi H}(j)$, one age profile from each equation. Picking the single number $\bar{\kappa}_{\phi d}$ settles both: the first equation turns it into $\kappa_{\phi d}(j)$ at every age, and the second then reads $\kappa_{\phi H}(j)$ off Baggett's ratio at every age.

Undoing the mixture needs the SUD's own multiplier at each age. That is what $\sigma_d(j)$ supplies: the age profile of drug poisoning deaths from CDC WONDER 2023, normalised to one at the age where it peaks, so $\sigma_d(j)$ is the share of the SUD's peak mortality effect that is present at age j. Writing $\bar{\kappa}_{\phi d}$ for the level of the SUD multiplier,

$$\kappa_{\phi d}(j) = 1 + (\bar{\kappa}_{\phi d} - 1)\,\sigma_d(j), \qquad \kappa_{\phi H}(j) = \max\Big\{1,\ \frac{RR(j)}{(1-s) + s\,\kappa_{\phi d}(j)}\Big\}. \tag{25}$$

The denominator is the average mortality multiplier in Baggett's population: a share $1-s$ of it carries homelessness alone and a share s carries the SUD as well. Dividing the observed ratio by it leaves the homelessness effect on its own.

$\bar{\kappa}_{\phi d}$ is assigned from the literature on all cause standardised mortality ratios for opioid use disorder, which report values between 6 and 10 (Degenhardt et al., 2011; Hser et al., 2017; Larney et al., 2020), and the calibration section reports the value taken. The drug price participation elasticity is not targeted; it is reported as a validation moment and compared with the range estimated in the literature, -0.3 to -1.1 (Saffer and Chaloupka, 1999; Grossman and Chaloupka, 1998; Olmstead et al., 2015).

$\kappa_{\phi m}$ carries no age argument because the data give it basically none. Olfson et al. (2015) give all cause standardised mortality ratios for schizophrenia of 3.6, 3.4 and 4.0 across ages 20 to 34, 35 to 54 and 55 to 64. Flat, if anything rising.

Unemployment risk. Job finding and separation are

$$\pi^{ue} = \min\Big\{1,\ \bar{\pi}^{ue}\,\kappa_{ueH}^{\mathbf{1}\{h'=h_1\}}\kappa_{ued}^{\mathbf{1}\{d=1\}}\kappa_{uem}^{\mathbf{1}\{m=1\}}\Big\}, \qquad \pi^{eu} = \min\Big\{1,\ \bar{\pi}^{eu}\,\kappa_{ueH}^{-\mathbf{1}\{h'=h_1\}}\kappa_{ued}^{-\mathbf{1}\{d=1\}}\kappa_{uem}^{-\mathbf{1}\{m=1\}}\Big\}, \tag{26}$$

with $\bar{\pi}^{ue}$ and $\bar{\pi}^{eu}$ the baseline job finding and separation rates and κ_{ueH}, κ_{ued} and κ_{uem} the multipliers for homelessness, the SUD and SMI, so each multiplier below one makes work simultaneously harder to find and easier to lose.

The child's income risk. A parent with a SUD, or one who ends up on the street, lowers the productivity (y_{child}) its child is born with, through δ_d and δ_h respectively, as shown in (39) in Section 2.14. That is a risk

to the household's own objective only to the extent that ω, the weight the parent places on its child's lifetime utility in (30), is positive. Since ω is small (see Section 3 for details), the household weighs a fraction ω of this risk and $1-\omega$ of it falls outside the problem entirely. It is therefore the one risk here that is principally an externality rather than a pure personal cost, which is what makes it a candidate for policy in a way that mortality and unemployment risk, both fully internalised, are not.

2.9 The voucher, and the two policies

A household without a voucher draws one with probability π_V, the same lottery probability as in Section 1.2.3, if it is income eligible, and a household with one keeps it while eligible, so unlike Section 1.2.3 the voucher is persistent rather than redrawn each period. Eligibility is

$$y^g + ra \;<\; \kappa_V \, y^{\text{med}}. \tag{27}$$

with κ_V the eligibility cutoff expressed as a share of median gross income y^{med}. Under Housing First that is the whole rule. Under Treatment First the voucher is additionally lost with probability p_{det} by a household that uses drugs or has a SUD, and with probability p_{fp} by one that did not,

$$\Pr(\eta'=1) = \Pr(\eta'=1)^{HF} \times \begin{cases} 1-p_{\text{fp}} & \text{if } (v=0 \text{ and } d=0) \text{ or } \tau=1, \\ 1-p_{\text{det}} & \text{otherwise.} \end{cases} \tag{28}$$

There are four cases and the rule covers them. A clean household that does not use meets the condition and faces only the false positive rate p_{fp}. A clean household that uses fails it and faces p_{det}. A household carrying the disorder that does not take treatment fails it and faces p_{det} as well. And a household carrying the disorder that does take treatment is abstaining, $v=0$ by the restriction in Section 2.6, so it meets the condition and faces p_{fp}. There is no fifth case: treatment is available only when $d=1$, and it forbids use, so no household is ever both in treatment and using.

This is the one place where the quantitative model changes the *logic* of Section 1.5 rather than its setting. There, conditionality is perfectly enforced and perfectly met, so Treatment First deters everyone and nobody is thrown off the rolls. Here detection is imperfect and so are households: some fail the condition because they cannot get clean, and some fail it because the test is wrong. Those two groups are the empirical case for Housing First, neither exists in the stylized model, and that is why its welfare numbers were illustration.

2.10 The shocks, collected

Conditional on the state and the choices, next period's state is drawn from five independent transitions. Collecting them in one place:

Productivity y. Fixed over the life cycle, so $y'=y$. I do this for two reasons. First, it is what keeps the child's penalty (39) a shift along a single scalar rather than a shift across the nodes of an age varying discretisation, which significantly reduces the computation time to manageable levels. Second, it still captures the key mechanisms, because unemployment and the penalty ρ_d are the income risks that link the SUD to housing, and those are retained in full.

Employment e. A household that loses their job receives insurance with certainty for one year. With π^{ue} and π^{eu} from (26), it gives the following transitions:

$$e=1 \;\longrightarrow \begin{cases} e'=1 & \text{w.p. } 1-\pi^{eu} \\ e'=2 & \text{w.p. } \pi^{eu} \end{cases} \qquad e\in\{2,3\} \;\longrightarrow \begin{cases} e'=1 & \text{w.p. } \pi^{ue} \\ e'=3 & \text{w.p. } 1-\pi^{ue}, \end{cases} \tag{29}$$

State 2 is reachable only from employment and never from itself, so a household is in it for the first year out of work and in state 3, which pays nothing, from the second year on. One year is if anything generous: most states pay for twenty six weeks.

The SUD d. By (20) if $d = 0$ and (21) if $d = 1$, conditioning on the chosen v, h' and τ.

Mental illness m. By (23), conditioning on d.

The voucher η. By (27) and (28), conditioning on next period's eligibility and, under Treatment First, on the chosen v, the status d and the treatment choice τ.

Write $\Gamma\big(s' \mid s,\, n, h', v, \tau\big)$ for the product of these five, which is the object the expectation in (30) is taken against.

2.11 The household's dynamic problem

Putting it together, for $j = 1, \ldots, J$ and with $V(J{+}1, \cdot)$ given below,

$$
\begin{aligned}
V(j,a,y,e,d,m,\eta) = \max_{\{n,\,h',\,v,\,\tau,\,a'\}} \Bigg\{ & \frac{\big[c^{1-\gamma-\theta}(h')^{\gamma}(v+b)^{\theta}\big]^{1-\sigma}}{1-\sigma} + \omega_{\ell}\frac{(1-n)^{1-\sigma_{\ell}}}{1-\sigma_{\ell}} + \aleph - \tau\bar{\tau}^{u} - \kappa_x H \\
& + \ \beta\,\phi(j,h',d,m)\int_{\mathcal{S}} V(j{+}1)(s')\,\Gamma\big(\mathrm{d}s' \mid s,n,h',v,\tau\big) \\
& + \ \beta\,\big[1-\phi(j,h',d,m)\big]\,\omega\,\mathbb{E}\,V\big(1,\,0,\,y_{\text{child}},\,e',\,0,0,0\big) \Bigg\}
\end{aligned}
\tag{30}
$$

subject to the budget (18), the income definitions (15) to (17), the choice set of dwellings (13), $\tau = 0$ whenever $d = 0$, $n = 0$ whenever $j \geq J_R$ or $e \neq 1$, $v = 0$ whenever $\tau = 1$, $a' \geq 0$, $c \geq F$ so that consumption never falls below the food stamp floor, and $V(J{+}1, \cdot) = 0$ with $\phi(J) = 0$, so that at the last age the child is the whole continuation. The integral runs over next period's state $s' = (a', y, e', d', m', \eta')$; the second line of (30) is the household's own continuation, and the third is the child's expected lifetime utility, entered at age one with zero assets and with its own employment status e' drawn as any newborn's is.

A household that does not survive from j to $j + 1$ leaves a child, so the continuation is the survival weighted value of its own next period plus the death weighted value of the child's life, each discounted, with y_{child} as defined in (39). What that buys is a second route by which the household's own decisions, to use and to go unhoused, reach its child. $\phi(j, \cdot)$ itself depends on h' and, through d, on v: a household that uses raises the chance it is the child's term rather than its own that is realised, so the child enters the drug and housing decision through the weight on the two terms as well as through y_{child}.

At $\omega = 0$ death is worth zero and the damage a household does to its child is entirely external; at $\omega > 0$ the household internalises a share ω of it, and only $1 - \omega$ remains external. Since the child's productivity depends on whether the parent carried a SUD or was homeless, and both depend on v and h', the child term in (30) is what puts the child inside the parent's drug and housing decision.

How the choices separate, and where they do not. Taking hours, the dwelling and next period assets as given, what is left to divide between consumption and the drug is

$$
x \;=\; y^{n} + a \;-\; p_r\, h'\, \mathbf{1}\{h' \neq h_1\} \;+\; Vch + F \;-\; a', \tag{31}
$$

so the budget (18) reads $c + p_v v = x$ and the whole of the remaining problem is one number: how to split x between c and v, since either one determines the other.

Were the drug an ordinary good, one whose only consequence is the flow utility it delivers this period, that split would follow from flow utility alone. It does not, because v enters the transition as well as the utility: by (20) another unit of the drug also buys a higher chance of waking up next period with the disorder. So there are three terms in the first order condition for v, not two,

$$
\underbrace{\frac{\theta}{b+v}\,\Lambda}_{\text{value of another unit}} \;=\; \underbrace{\frac{p_v\,(1-\gamma-\theta)}{c}\,\Lambda}_{\text{the consumption it displaces}} \;+\; \underbrace{\lambda e^{-\lambda v}\Big[\,\mathbb{E}V(\cdot \mid d'=0) - \mathbb{E}V(\cdot \mid d'=1)\,\Big]}_{\text{the risk it adds}}, \tag{32}
$$

where $\Lambda = \left[c^{1-\gamma-\theta}(h')^{\gamma}(v+b)^{\theta}\right]^{1-\sigma}$ is the Cobb-Douglas bundle of (14). It multiplies the first two terms and not the third, because the third is a difference of continuation values and is therefore already in utils. Were it absent, Λ would divide out of the condition entirely and the split would reduce to the usual Cobb-Douglas one; it is the third term that stops it cancelling, and that is why the marginal utility level matters here and not in an ordinary two good problem. That third term is the marginal hazard $\pi_d'(v) = \lambda e^{-\lambda v}$ times what the disorder costs, the gap between the two continuation values. This is Section 1's mechanism inside the quantitative model: because π_d rises in v, how much to use is not decided on the budget alone.

There are two cases in which the third term is zero, and in both the household faces no such trade off: one that abstains, where $v = 0$, and one that already carries the disorder, whose recovery hazard (21) does not depend on how much it uses. For both the condition is the Cobb-Douglas split in closed form,

$$c = \max\left\{F,\ \frac{1-\gamma-\theta}{1-\gamma}\big(x + p_v b\big)\right\}, \qquad v = \max\left\{0,\ \frac{x-c}{p_v}\right\}, \tag{33}$$

whose corner is the abstention decision: a household buys nothing when the Stone-Geary endowment already exceeds what it wants, which is how a large b deters use and is the threshold of Section 1.2.4. The outer maximum on c is the food stamp floor. Where it binds the Cobb-Douglas split no longer describes the allocation: consumption is held at F and the drug takes what the budget leaves, so the floor changes behaviour and not only welfare. Where the third term is present the answer lies between (33) and abstention, since risk can only push towards less, so it is found by a one dimensional search on that interval.

Hours then solve

$$u_c(c)\,\frac{\partial x}{\partial n} \;=\; \omega_\ell (1-n)^{-\sigma_\ell}, \tag{34}$$

and the clawbacks in (17), at a rate of one for food stamps and κ_r for the voucher, make $\partial x/\partial n$ jump *up* at the hours where each transfer runs out, because the clawback stops. That is the welfare trap, and it is the channel through which conditionality changes labour supply rather than only consumption.

2.12 The government

The government pays social security, unemployment insurance, disability, food stamps and vouchers, and supplies a public good $\bar{G}$ that does not enter private consumption and is held fixed across simulations, so that policies are compared at a constant level of it. It finances this with the two taxes in (16) and the estates of the deceased, with the level τ_ℓ clearing the budget in every simulation,

$$\underbrace{\textstyle\sum y^g - \tau_\ell \sum (y^g)^{\xi} + \tau_k \sum ra + \text{estates}}_{\text{revenue}} \;=\; \underbrace{SS + UI + DI + F + Vch + \bar{G}}_{\text{outlay}}. \tag{35}$$

In the stylized model the two policies happen to cost the same and the comparison is unaffected. Here a policy that houses more households spends more, τ_ℓ rises to pay for it, and the higher tax lowers the after tax return to an extra hour: it enters (34) through $\partial x/\partial n$, which is net of the tax schedule in (16), so hours fall and saving with them, for everyone, including the households the programme never reaches, and it runs against Housing First.

2.13 The drug market

Drug production has cost $C(S_v) = \frac{\nu_1}{\nu_2} S_v^{\nu_2}$, where S_v is the aggregate quantity. Supply satisfies

$$p_v = \nu_1 S_v^{\nu_2 - 1}, \qquad S_v = \int v(s)\,\mathrm{d}\Phi(s), \tag{36}$$

with price elasticity of supply $1/(\nu_2 - 1)$ and ν_1 normalised so that the quantity supplied equals the quantity demanded at the benchmark price, which is normalised to one. The supply elasticity, its range and the evidence behind them are set in Section 3.

2.14 The law of motion and newborns

Given the policy functions, the distribution over states evolves by pushing households through their choices and then through the shocks,

$$\Phi_{j+1}(\mathcal{S}') \;=\; \phi(j,\cdot)\int \Gamma\big(\mathcal{S}' \mid s,\, n_j(s), h'_j(s), v_j(s), \tau_j(s)\big)\,\mathrm{d}\Phi_j(s) \qquad \text{for every measurable } \mathcal{S}', \tag{37}$$

together with the newborn inflow

$$\Phi_1(\mathcal{S}') \;=\; \int \big[1-\phi(j,\cdot)\big]\,\mathcal{N}\big(\mathcal{S}' \mid s\big)\,\mathrm{d}\Phi_j(s) \;+\; \int \mathcal{N}\big(\mathcal{S}' \mid s\big)\,\mathrm{d}\Phi_J(s), \tag{38}$$

where $\mathcal{N}$ places a newborn according to (39), so that the mass of newborns equals the mass of the dead exactly and the two equations together conserve the population.

Here $\phi(j,\cdot)$ is evaluated at the household's own $(h'(j,s), d, m)$. The mass that does not survive, plus the mass at $j = J$, is replaced by newborns at $j = 1$ with zero assets, clean, without SMI and without a voucher, and subject to the same unemployment shocks as anyone else, whose productivity is drawn from the unconditional distribution Ω_y of permanent productivity across households scaled by the transmission penalties δ_d for a child whose parent carried the SUD and δ_h for one whose parent was homeless,

$$y_{\text{child}} \;\sim\; \Omega_y \;\text{ scaled by }\; (1-\delta_d)^{\mathbf{1}\{\text{parent had the SUD}\}}\,(1-\delta_h)^{\mathbf{1}\{\text{parent was homeless}\}}. \tag{39}$$

2.15 Competitive equilibrium

A competitive equilibrium is a collection of

- value functions $\big\{\{V_t(j,\cdot)\}_{j=1}^{J}\big\}_{t=0}^{\infty}$;
- policy functions $\big\{\{n_t(j,\cdot),\, h'_t(j,\cdot),\, v_t(j,\cdot),\, \tau_t(j,\cdot),\, a'_t(j,\cdot)\}_{j=1}^{J}\big\}_{t=0}^{\infty}$;
- distributions $\big\{\{\Phi_t(j,\cdot)\}_{j=1}^{J}\big\}_{t=0}^{\infty}$; and
- prices, policies and aggregates $\big\{\tau_{\ell t},\, p_{vt},\, \bar{y}_t,\, y_t^{\text{med}},\, H_t\big\}_{t=0}^{\infty}$,

such that, at every date t,

1. the policy functions solve (30) given its terminal condition, and $\{V_t(j,\cdot)\}_{j=1}^{J}$ are the associated values;
2. $\Phi_{t+1}(j{+}1,\cdot)$ follows from $\Phi_t(j,\cdot)$ by the law of motion (37) together with the newborn rule (39) and (38), and the mass of newborns equals the mass of the deceased;
3. the government budget (35) holds at $\tau_{\ell t}$;
4. the drug market clears at p_{vt}, so (36) holds with aggregate drug demand $S_{vt} = \int v_t(s)\,\mathrm{d}\Phi_t$;
5. the public good is supplied at its assigned level, $G_t = \bar{G}$;
6. $\bar{y}_t$, median gross income y_t^{med} and the homelessness rate H_t are the aggregates implied by Φ_t.

A stationary equilibrium is the special case in which all of these are constant and $\Phi_{t+1} = \Phi_t$, which is what the calibration and the policy comparisons report. The sequence version is stated for completeness, and because the transition exercise in the results needs it.

3 Calibrating the quantitative model

The model has fifty four parameters, some of which vary in value by age. Thirty two are calibrated externally, from data or from the literature, and twenty two are calibrated internally, inside the model. Four further parameters govern drug testing and detection; they are inert in the benchmark, where nothing is conditional, so they are set with the Treatment First experiment rather than here. They are split across the two tables of this section: thirty two are the main story of this paper and its policy analysis (Table 5, seventeen internally and fifteen externally calibrated), and twenty two are standard (Table 6, five internally and seventeen externally).

The tables carry the parameter, its description, the moment and source used to fix its value, and the value itself. The text alongside each table adds what a table has no room for: the intuition for what each parameter does, and why its moment identifies it.

3.1 The thirty two parameters that carry the story of this paper

Most of the fifty four are the furniture of any life cycle model. A much smaller set is specific to this paper, and all thirty two of those carry its main story and policy analysis. They are set out in Table 5, with the parameter, its description, and the moment and source used to fix its value, which is given in the last column. They are discussed one at a time below because they are the parameters a reader may want to argue over, and each is taken in the order of the table.

The comparison has a shape, and every entry below is read against it, so it is worth setting out once and in full.

Two objects decide the ranking. The **private value** is what a household itself gains from being housed and from not carrying a disorder: consumption, a dwelling rather than the street, and above all the years of life that homelessness and the disorder take away. The **externality** is everything that lands on somebody else: on the child, who inherits not the disorder but a worse productivity draw, and so is likelier in turn to be poor, to use, to acquire a disorder and mental illness as a consequence, to be unhoused and to die young; on taxpayers, who fund both programmes and lose hours to the tax; and on everybody, through the disamenity of aggregate homelessness. The parent internalises only ω of the part that falls on its child, and ω is small, which is what leaves anything for a policy to correct.

Housing First is redistribution, and it removes externalities while creating another. It moves resources to the poorest households and spares them the outcomes that fall hardest on them, which is most of its case. Mechanically it also removes two externalities: a child of a housed parent does not carry the penalty δ_h prices, and aggregate homelessness falls, which every household values through κ_x. It creates one at the same time, because assistance that arrives whatever a household does insures it against the consequences of using, so the SUD rate rises and reaches the child through δ_d.

Treatment First is the mirror image, and it removes externalities but creates another too. Curing the parent is how the child's worse draw is avoided, which is its case. But conditioning assistance means some households do not receive it, so Treatment First mechanically raises homelessness, especially in the short run, and over that horizon it creates the two externalities Housing First removes. The short run matters on its own account: a household that wants to stop cannot do it at once, and the transition is costly while it lasts, which is why the policy analysis reports the transition and not only the steady states.

Both are paid for. The labour income tax funds the vouchers under Housing First and the monitoring under Treatment First, and in either case it lowers the after tax return to an hour for households the programme never reaches, while the claw back at rate κ_r does the same for recipients. The channel also runs back: a household that uses is likelier to be out of work, so it pays less tax and draws more transfers, which requires even higher taxes on everybody else.

And general equilibrium returns part of it. The drug price is endogenous, set by the supply curve ν_1 and ν_2 describe, so curing disorders lowers drug demand, lowers the price, and makes using cheaper for those still deciding: part of what treatment removes comes back through the market it operates in. Spend more on either policy and the tax rises, hours fall, the base shrinks and the tax has to rise again. Treatment First has an offset of its own in the other direction: sending more households into treatment raises the recovery rate faced by everyone deciding whether to use, and a newborn takes those consequences into account when it makes its own decision, so the policy changes behaviour beyond the households it treats. And because κ_x is levied on the aggregate headcount, any movement in homelessness feeds back into everyone's utility. So the mechanical effects above are upper bounds on both sides.

That is the whole apparatus, and it is why each entry below can be short. For every parameter the two things worth knowing are what happens to the private value and what happens to the externality *as it rises*, and each entry says so in a line before giving the moment. The middle blocks are where the two move together: because the child inherits a higher chance of landing in exactly the states those parameters price, anything that raises the private cost of using or of being unhoused raises the externality as well. Both policies work

on those blocks, from opposite ends.

Table 5: The thirty two parameters that carry the story of this paper

Parameter	Description	Set	Moment and source	Value
Who uses drugs				
θ	Weight on drugs in utility	Int.	Drug spending given a SUD, NSDUH 2024	—
b	Whether a household uses	Int.	Drug participation, NSDUH 2024	—
λ	SUD hazard per unit used	Int.	Users acquiring a SUD, NSDUH 2024	—
$\bar{\kappa}_{\phi d}$	Mortality with a SUD	Ext.	Mortality ratios, Hjorthøj et al. (2023)	8.00
$\aleph$	Flow value of being alive	Int.	Value of a statistical life, Viscusi (2003)	—
What using and being homeless cost				
ρ_d	Earnings lost to a SUD	Int.	Employment rate of the homeless, HUD 2023	—
κ_{ued}	Employment multiplier, SUD	Int.	Unemployment rate given a SUD, NSDUH 2024	—
h_1	Service flow of homelessness	Int.	Homelessness rate, HUD 2023	—
$\kappa_{\phi H}(j)$	Increased mortality when homeless	Ext.	Mortality ratios by age, Baggett et al. (2013)	4.41 to 1.09
κ_{ueH}	Employment multiplier, homeless	Ext.	Desmond and Gershenson (2016)	0.80
ζ_{DI}	Disability replacement, SMI	Ext.	SSDI and SSI rates, SSA 2023	0.25
$\underline{c}$	Food stamp consumption floor	Int.	Food stamps over labour income, USDA	—
π_V	Voucher lottery probability	Int.	Recipients over eligible, HUD 2023	—
The mental illness chain				
p_m	Prevalence the onset reproduces	Ext.	Share with SMI, NSDUH 2023	5.5%
λ_m	Annual chance of leaving SMI	Int.	SMI share of homeless, Gutwinski et al. (2021)	—
ι	SUD multiplier on SMI onset	Ext.	Danish registers, Nielsen et al. (2017)	2.00
ρ_m	Earnings lost to SMI	Int.	Earnings of workers with SMI, NSDUH 2023	—
κ_{uem}	Employment multiplier, SMI	Int.	Employment rate given SMI, NSDUH 2023	—
μ_m	SUD recovery multiplier, SMI	Ext.	Dual diagnosis relapse, Bradizza et al. (2006)	0.82
$\kappa_{\phi m}$	Mortality multiplier, SMI	Ext.	Olfson et al. (2015)	2.50
Recovery and treatment				
$\bar{\pi}_r$	Baseline SUD recovery rate	Ext.	Ten year remission rate, Vaillant (1995)	0.025
μ_H	Recovery multiplier when homeless	Int.	Weeks abstinent, Milby et al. (2005)	—
g	Treatment effect on recovery	Ext.	Odds ratio, Milby et al. (2005)	1.35
$\bar{\tau}^u$	Utility cost of treatment	Int.	Treatment take up, NSDUH 2023	—
The externalities				
ω	Weight on the child	Int.	Parent use rate with a child, NSDUH 2024	—
δ_d	Child's earnings, parent SUD	Int.	Child SUD ratio, McGovern et al. (2023)	—
δ_h	Child's earnings, homeless	Ext.	Poverty exposure on child's earnings, Chetty et al. (2016)	0.31
ω_ℓ	Weight on leisure	Int.	Average hours worked, Guvenen et al. (2023)	—
κ_x	Disamenity of homelessness	Ext.	Willingness to pay to eliminate homelessness, Loubière et al. (2020)	0.0008
The drug market				
ν_1	Drug supply shifter	Ext.	Normalisation, so $p_v = 1$ in the benchmark	norm.
p_v	Price of drugs	Ext.	Clears the drug market in the benchmark	1.00
ν_2	Drug supply elasticity	Ext.	Elasticity 2; Hansen (2017), Chaloupka (1991)	1.50

Who uses drugs. These four set participation and, more importantly, *who* participates. Neither policy acts on them directly; they decide how much there is for a policy to act on.

θ, the weight on the drug in utility. How much a household wants the drug at all, against consumption and housing. It is the level of the temptation where b is its curvature, and it is the parameter the amount spent on drugs turns on. As it rises use rises on both margins, so the private cost and the externality both rise and there is more for either policy to act on. It is calibrated internally to drug spending given a SUD, NSDUH 2024.

b, whether a household uses at all. The marginal value of the drug goes as $\theta/(b+v)$, so a larger b makes the first units worth less and a household with little to spend buys nothing. As b rises use falls, so the private cost and the externality both fall, and a transfer is less likely to be spent on drugs: b is the size of the moral hazard. Its value is chosen so the model generates a drug participation rate of 5.13 percent, the NSDUH 2024 figure for adults aged 26 and over, which identifies it sharply because participation is close to a threshold function of b.

λ, the hazard per unit used. This is what makes using *more* genuinely riskier rather than merely dearer, and it is the step that turns spending into a disorder. As λ rises both the private cost and the externality rise, and Housing First's moral hazard costs more, since it is λ that converts the extra a voucher buys into the SUD rate Treatment First attempts to cure. Its value is chosen so the model generates the share of users who go on to acquire a SUD, from the same NSDUH 2024 sample. The flow is the

moment rather than the stock because prevalence cannot separate onset from recovery, and recovery is set separately below, which is what leaves this moment free to identify onset.

$\bar{\kappa}_{\phi d}$**, mortality with a SUD.** This is the parameter the whole paper turns on. Mortality truncates the value of life for a user, and hardest for households with the most life left to value, which is what reverses the income gradient and makes the poor the users rather than the rich. As it rises the private cost rises sharply, because it is years of life; the externality rises with it, since a parent who dies hands the draw on; and use falls, which cuts both. Both policies reduce these deaths, treatment by curing and housing because recovery is less likely when homeless. It is calibrated externally, at the midpoint of the standardised mortality ratios for SUD in Hjorthøj et al. (2023), which run six to ten, with the age profile from CDC WONDER 2023. Setting it externally is what frees the drug price elasticity to test the mechanism rather than pin a parameter.

$\aleph$**, the flow value of being alive.** The other half of the mortality mechanism: $\bar{\kappa}_{\phi d}$ sets how much a SUD raises the chance of dying and $\aleph$ sets what the life being lost is worth. As it rises the private value of every life either policy saves rises, and use falls, so the externality falls. It needs a level and not only a behavioural target: at a negative value households prefer death and the mortality cost of using becomes a benefit. Its value is chosen so the model generates a value of a statistical life of \$7.5m, from Viscusi and Aldy (2003).

What using and being homeless cost, and the voucher. Both policies work on this block, from opposite ends: each parameter says how bad a state is, so each sets what a voucher is worth to a household in it and what being cured of the reason for it is worth. The voucher itself closes the block, since it is the instrument that acts on all of them. Because the child inherits a higher chance of landing in these states, everything here moves the private value and the externality together.

ρ_d**, earnings lost to a SUD while still working.** It reduces the value of life when a household acquires a SUD, since the earnings it costs would have been enjoyed for the rest of that life. It also does a job nothing else can: it lets an *employed* household be poor enough to be priced out of housing, which an employment effect alone cannot deliver, since that makes every homeless household jobless and about half are not. As ρ_d rises the penalty shrinks, so the private cost falls and, because the child's draw improves with it, so does the externality. These earnings are lost housed or homeless, so treatment recovers them and housing does not. One subtlety is worth stating, because it may seem like a contradiction with Section 1.1. There the proportional income penalty was shown to matter little for *who* uses. That is a statement about the calibrated model, not the parameter: raising ρ_d with b held fixed does change who uses, but b is recalibrated to the participation rate each time the penalty changes, and once it is, the composition of users is back where it started. Its value is chosen so the model generates the employment rate of the homeless, about half, from HUD's 2023 point-in-time count and TEDS-A 2022, which is its moment because it is the parameter that puts working households on the street.

κ_{ued}**, the employment multiplier for a SUD.** The same channel on the extensive margin: regaining employment is less likely and losing it more likely with a disorder. As it rises the private cost rises, the externality rises with the child's draw, and the tax base narrows, so it costs taxpayers as well. The two penalties are told apart by the moments they move: ρ_d shifts income *among the employed*, which the homeless employment rate reads, while κ_{ued} shifts the employment rate itself. Its value is chosen so the model generates the unemployment rate given a SUD, 16.25 percent in NSDUH 2024.

h_1**, the service flow of homelessness.** It sets how bad the outside option is, and the value of being alive is measured against that outside option: a household with little to lose from dying is one for which the mortality cost of using is small. As h_1 rises the street is less bad, so the private cost of homelessness falls, a voucher is worth less, and the externality on a child likelier to be unhoused falls with it. It is therefore what Housing First's redistribution is worth. Its value is chosen so the model generates the homelessness rate, 0.399 percent of households in HUD's 2023 point-in-time count.

$\kappa_{\phi H}(j)$**, mortality when homeless.** As it rises the private cost of being unhoused rises and so does the private value of a voucher, and the externality rises at the same time, because the parents who die are parents of children and the workers who die were paying tax. Baggett et al. (2013) give race adjusted

mortality ratios for homeless adults of about 9.1 at ages 25 to 44, falling to 1.1 at 65 to 84. Those are totals for a population most of which also carries a SUD, so it is set externally age by age net of that composition, which is why it is derived jointly with $\bar{\kappa}_{\phi d}$ rather than read off directly, as set out in Section 2.8 at (25).

κ_{ueH}**, employment lost to homelessness.** Here the homelessness itself costs the job, so housing mechanically recovers the earnings and treating does not. As it rises the private cost and the externality rise together, and so does what housing recovers, which makes it the clearest case in the table of a parameter loading Housing First's side. It is set externally at 0.80, from Desmond and Gershenson (2016).

ζ_{DI}**, the disability transfer for the non employed with SMI.** It is the floor under the worst state in the model, being ill and out of work, so it sets how far a household can fall. As it rises the private cost of that state falls, and the externality on a child likelier to reach it falls, at the price of a higher tax on everybody else. It is set externally from SSDI and SSI replacement rates, SSA 2023.

$\underline{c}$**, the food stamp consumption floor.** The floor under consumption, and it belongs here rather than among the standard parameters because it is what the value of life rests on at the bottom of the distribution: a household that cannot fall below $\underline{c}$ has something to lose from dying even with no assets and no job, and that is exactly the household the mortality channel decides. As it rises the private cost of every bad state falls, so a voucher is worth less at the margin and the externality on a child likelier to reach those states falls too, while the tax that funds it rises. It is calibrated internally to food stamp spending over labour income, from the USDA.

π_V**, the chance of drawing a voucher.** This is the policy dial itself. As it rises the private gain rises and two externalities fall mechanically, the child's penalty for growing up unhoused and the aggregate disamenity, while the third rises through the SUD rate and the tax rises with the bill: the whole trade in one parameter. Its value is chosen so the model generates the 26.6 percent ratio of voucher recipients to the income eligible population in HUD's Picture of Subsidized Households, 2023.

The mental illness chain. This block is why Treatment First reaches further than the disorder it treats: a SUD raises the chance of mental illness, mental illness lowers earnings and employment and raises death, and the child inherits all of it as a worse draw. Everything here therefore moves both objects in the same direction, and the first three decide whether the chain exists at all.

p_m**, the prevalence the onset reproduces.** The onset hazard is not free: it is backed out of the observed age profile of prevalence and the exit rate by the identity in (22), so the profile is reproduced by construction and the level of mental illness in the model is data rather than a fitted number. As it rises more households are in the state, so both objects rise. It is set externally from the share of adults with SMI in NSDUH 2023.

λ_m**, the annual chance of leaving SMI.** The persistence of the state, and the one piece of the chain the data cannot hand over, since prevalence pins the stock and not the flows. As it rises mental illness is less persistent and both objects fall. It is calibrated internally to the SMI share of the homeless in Gutwinski et al. (2021): only a sufficiently sticky state lets people with SMI accumulate in homelessness at the ratio observed.

ι**, the SUD multiplier on SMI onset.** The hinge of the whole block. Without it mental illness is exogenous to everything a policy does and treating the disorder cannot touch it. As it rises a SUD causes more mental illness, so both objects rise and Treatment First's return rises with them, because curing the disorder now prevents the illness too. It is set externally at two, from the Danish registers in Nielsen et al. (2017).

ρ_m**, earnings lost to SMI.** The counterpart of ρ_d for mental illness, and it reaches the comparison through ι: a household carrying the disorder faces onset at ι times the baseline rate, so treatment reduces mental illness in a roundabout way and recovers these earnings for households that never fall ill. As it rises the penalty shrinks and both objects fall. Its value is chosen so the model generates the earnings of workers with SMI relative to average, from NSDUH 2023.

κ_{uem}**, the employment multiplier for SMI.** The same on the extensive margin, and where the chain does most of its work, since employment is what mental illness costs most. As it rises both objects

rise and the tax base narrows; a child of a treated parent is less likely to carry SMI, which raises its earnings and lowers its chance of being unhoused, so part of Treatment First's return never appears in the treated household's own drug use. Its value is chosen so the model generates the employment rate given SMI, about 30 percent, the midpoint of the range for high disability diagnoses in NSDUH 2023.

μ_m**, the SUD recovery multiplier for SMI.** The second direction of the chain: not only does a SUD cause mental illness, mental illness keeps the SUD. As it falls the disorder is harder to leave, so both objects rise and treatment has more to overcome in exactly the households it most wants to reach. It is set externally at 0.82 from dual diagnosis relapse rates in Bradizza et al. (2006), and it is one of the three values in the paper carrying no real discipline, which Section 3.3 says.

$\kappa_{\phi m}$**, the mortality multiplier for SMI.** The chain's last link, and what makes it matter for welfare rather than only for income. As it rises deaths among the mentally ill rise, so both objects rise, and part of what a SUD does to mortality it does by way of SMI rather than directly. It is set externally at 2.5, the middle of the standardised mortality ratios for SMI in Olfson et al. (2015).

Recovery and treatment. This is the block Treatment First works on. The first two set how much recovery there is to buy and the last two how much of it a policy can actually buy.

$\bar{\pi}_r$**, the baseline SUD recovery rate.** The hazard of getting out of the disorder for a housed household, which is the level everything else in this block scales. As it rises disorders end sooner on their own, so both objects fall and there is less for either policy to buy. It is set externally from the ten year remission rate in Vaillant (1995).

μ_H**, SUD recovery when homeless.** How much less likely recovery from the disorder is for an unhoused household than a housed one. It matters more than the level $\bar{\pi}_r$ because it is a *gradient*: it is what a voucher buys beyond shelter. As it falls the gradient steepens, so housing buys more recovery: the private value of a voucher rises and the externality falls, which is what offsets the moral hazard in π_V. Near one, assistance is a transfer and the moral hazard stands unopposed. Its value is chosen so the model generates the increase in recovery housing produced in Milby et al. (2005), a randomised trial on cocaine dependent homeless adults: 6.60 weeks abstinent in the housed arm against 4.48 homeless, so housing raises recovery by 47.3 percent.

g**, the treatment effect on recovery.** Treatment First's instrument, and the parameter its case rests on most directly. As it rises treatment works better, so the externality falls and Treatment First's case strengthens; the private cost falls with it, since the household is cured too. It is set externally from the odds ratio in Milby et al. (2005), and because it carries a headline result it is reported across a range rather than at a point.

$\bar{\tau}^u$**, the utility cost of treatment.** Take up is a choice, not a rule, and this is what makes households decline treatment they would gain from. As it rises fewer take treatment, so the externality falls by less than g alone would suggest and Treatment First delivers less than its instrument promises. It is calibrated internally to treatment take up in NSDUH 2023.

The externalities. The first three are what a parent's disorder and a parent's homelessness cost the child, and the last two act on households who are neither. This is where the ranking is decided, since the private comparison already favours Housing First.

ω**, the parent's weight on its child.** A parent values its child's lifetime utility at ω times its own, and ω is small. As it rises the parent already counts more of the child's loss, so the externality left for a policy to correct is smaller and both policies are worth less. Identifying it needs care, because *every* household here has a child, so there is no parent against non parent comparison to run inside the model. The moment is a counterfactual instead. In NSDUH 2024 past month use is 15.5 percent lower among parents than among non parents, with parents reweighted to the non parents' joint distribution of age, income, sex and marital status so the gap is behaviour rather than composition. Since ω is the only reason a household in the model uses less than it otherwise would, its value is chosen so that switching the child term off at the calibrated parameters raises drug use by that same 15.5 percent.

δ_d, **what the child loses to a parent's SUD.** This is Treatment First's case, very nearly in full. The child does not inherit the disorder, only a worse productivity draw, and the draw is what makes it likelier to be poor, to use, to acquire a disorder and mental illness in consequence, and to be unhoused. As it rises the externality rises and Treatment First's case with it, while the parent's own private cost is unchanged, which is exactly what makes it an externality. No paper estimates the effect of a parent's SUD on a child's lifetime earnings directly, which is why the penalty is calibrated to the disorder ratio rather than lifted from an estimate. What the literature supplies is the two halves of the chain: Felitti et al. (1998) link household dysfunction to adult outcomes, and Chetty et al. (2016) measure what childhood environment does to adult earnings. It is calibrated internally so the model reproduces the ratio of child disorder rates between parents with and without a disorder in McGovern et al. (2023), who find children of a parent with a SUD two to four times likelier to acquire one.

δ_h, **what the child loses to homelessness.** The externality Housing First removes, working through the same compounding draw as δ_d. As it rises the externality Housing First removes grows, again leaving the private cost unchanged. Chetty et al. (2016) find that moving a child to a lower poverty neighbourhood before age thirteen raises adult earnings by 31 percent; δ_h is calibrated externally to that figure, taking the earnings gain from leaving poverty as the mirror of the loss from growing up without stable housing.

ω_ℓ, **the weight on leisure.** This sets how much labour supply responds to the tax and to the programme's own design. As it rises hours are more responsive, so the tax that funds the vouchers or the monitoring costs more, and the distortion side of the comparison grows against whichever policy costs more. The channel also runs back, which is what makes it an externality rather than only a cost: a household that uses is likelier to be out of work and to need assistance, so its choice raises the tax everybody else pays. Its value is chosen so the model generates average hours worked, from Guvenen et al. (2023).

κ_x, **the disamenity of homelessness.** The only term through which one household's housing status enters the utility of another who is not its own child, which is what makes the homelessness rate a genuine fixed point rather than accounting. As it rises the externality from the aggregate headcount grows, which loads whichever policy houses more people. Setting it high therefore favours Housing First, so the higher figure is taken as the baseline and a ranking that still favours Treatment First survives the assumption least kind to it. It is calibrated externally from the contingent valuation in Loubière et al. (2020): €57 a year among the 42 percent of respondents willing to pay anything, which is 0.20 percent of average annual earnings, against €23 and 0.08 percent averaged over all respondents.

The drug market. Three parameters describe the supply of the drug. Two are normalisations and the third is a supply elasticity, and they matter for one thing: they make the drug price endogenous, so a policy that changes the number of users changes what the drug costs. That is the general equilibrium offset named above, and it exists only because these are here.

ν_1, **the supply shifter.** A pure normalisation, chosen so that supply equals demand at $p_v = 1$ in the benchmark. It normalises units, not behaviour. It is a normalisation, so nothing turns on its level: it is chosen so the benchmark price is one, and moving it moves neither object.

p_v, **the drug price.** One at the benchmark by that normalisation, and free in every counterfactual. That is what lets a policy which changes how much is used move the price, and so feed back into every other household's decision. As it rises using is dearer, so use falls and both objects fall with it. It is an equilibrium object rather than a dial, which is what lets a policy that changes the number of users change what the drug costs.

ν_2, **the supply elasticity.** Sets the price elasticity of supply at 2, the middle of a range from about 0.7 to about 4. There is no direct estimate for an illicit drug, so it is bracketed by the closest legal markets: the Washington State marijuana reform implies roughly 0.79, the cigarette and gasoline literatures nearer 3.6. As it rises supply is more elastic, so the price moves less when demand does, and the general equilibrium offset to treatment shrinks: at a very elastic supply, curing households barely lowers the price and almost none of the gain comes back.

3.2 Standard parameters

The rest are conventional, recorded so the calibration can be reproduced rather than because anything turns on them. Table 6 carries each with its source and vintage, in this order.

They are not inert in the experiments, and it is worth saying so plainly. The discount factor sets how heavily a death thirty years away is weighed, the dispersion of permanent income decides how many households are poor enough to be at risk, and risk aversion decides what insurance is worth: each of them moves the private value and, through the child's draw, the externality as well. What separates them from the thirty two above is not that they do nothing but that they are not specific to this paper. They are calibrated the way any life cycle model calibrates them, to the same moments and from the same sources, so a reader who disagrees with one of them is disagreeing with a literature rather than with anything argued here. The parameters in Section 3 are the ones where this paper makes a choice.

Table 6: Standard parameters

Parameter	Description	Set	Moment and source	Value
γ	CD weight on housing	Int.	Housing cost over income, SCF 2022	—
β	Discount factor	Int.	Wealth over output, BEA	—
$\underline{h}$	Smallest rental size	Int.	Housing cost over half income, SCF 2022	—
ς	Geometric spread of rentals	Int.	Housing cost over 30 pct, SCF 2022	—
$\bar{\pi}^{eu}$	Job separation rate	Int.	Unemployment rate, BLS CPS 2014–24	—
σ	Risk aversion	Ext.	Attanasio and Weber (2010)	2.00
σ_ℓ	Curvature on leisure	Ext.	Frisch elasticity of 0.5, Chetty (2012)	3.00
$\varepsilon(j)$	Age profile of efficiency	Ext.	Quadratic in age, PSID and CPS hump	profile
$\bar{\phi}(j)$	Baseline survival by age	Ext.	SSA period life table, ages 26 to 85	profile
ρ_y	Persistence of log prod.	Ext.	Storesletten et al. (2004)	0.90
σ_y	Dispersion of log prod.	Ext.	Storesletten et al. (2004)	0.20
j_E	Maximum experimentation age	Ext.	Reported with and without it	55
r	Risk free rate	Ext.	Kaplan, Mitman and Violante (2020)	0.03
ξ	Tax progressivity	Ext.	Heathcote et al. (2017)	0.85
τ_k	Capital income tax	Ext.	US long run rate on capital gains and	0.15
τ_ℓ	Tax level	Ext.	Clears the government budget	in eqm.
G	Other public spending	Ext.	Residual at benchmark, then held fixed	in eqm.
ψ	UI replacement rate	Ext.	DOL statutory replacement rates	0.50
$\bar{y}$	Cap on insurance earnings	Ext.	DOL state benefit maxima	1.00
κ_V	Voucher eligibility cutoff	Ext.	HUD rule	0.50
κ_r	Tenant contribution rate	Ext.	HUD rule	0.30
$\bar{\pi}^{ue}$	Job finding rate, annual	Ext.	Krueger, Cramer and Cho (2014)	0.55

β, σ, σ_ℓ. Discount factor, value chosen to match wealth over output; risk aversion and the curvature on leisure, both standard.

$\underline{h}$ and ς. The rental ladder, which is geometric, so these two numbers give the whole grid and no size is set by hand. They are matched to the shares of renters spending more than half and more than thirty percent of income on housing, which separates a level from a spread. A uniform grid is a restriction on ς, not a neutral choice, and fits those two shares worse.

$\bar{\pi}^{eu}$ and $\bar{\pi}^{ue}$. Baseline separation and job finding rates.

$\varepsilon(j)$, $\bar{\phi}(j)$, ρ_y, σ_y, j_E. Age profiles of earnings and survival, the persistence and dispersion of permanent productivity, and the age at which use can begin.

r, ξ, τ_k, τ_ℓ, G. The interest rate and the tax system, with τ_ℓ not set at all but whatever clears the budget, and G held fixed across simulations.

ψ, $\bar{y}$, κ_V, κ_r. The replacement rate and cap for unemployment insurance and the eligibility and contribution rates for the voucher, all set by statute.

How these are solved for. Two points about the mechanics survive from the longer treatment this section used to carry, because nothing else says them.

Three moments are cliffs, not slopes. Most parameters move their moment smoothly, so a step in one direction tells you where to go next. Three do not, and treating them as though they did is how a calibration silently fails. Three moments are step like and are bisected rather than stepped, for the reason Section 1.3 gives: participation is a near threshold function of b, homelessness moves from 0.12 to 3.29 percent for a 35 percent change in h_1, and wealth over output is 5.46 at $\beta = 0.995$ and 0.38 at $\beta = 0.85$. All three are

bisected at the head of each sweep, in partial equilibrium, before anything else moves. The parameter to moment map is close to triangular and the sign of each parameter's effect on each moment is declared for every pair, so a mistaken sign shows up as a moment walking away from its target rather than as a silent failure.

The second stage matches responses, not steady states. These are partial equilibrium cohort experiments. A trial of two hundred people does not move the tax rate or the drug price, so prices are held at benchmark; the household problem is solved again, because each arm changes the rules a forward looking household faces. Milby et al. (2005) randomised 196 cocaine dependent homeless adults into three arms on identical day treatment, and the targeted moment is the ratio of the unconditional voucher arm to no assistance, $6.60/4.48 = 1.473$ weeks abstinent, which is what identifies μ_H. The conditional against unconditional comparison is deliberately *not* targeted, because the source reports it as insignificant in the raw randomised comparison; calibrating to it would repeat the error this revision is correcting elsewhere. The drug price participation elasticity, -0.63, is no longer targeted. It is reported below as a moment the model is not fitted to, which is the stronger use of it: with $\bar{\kappa}_{\phi d}$ assigned from mortality data, the elasticity tests whether the mechanism delivers the observed price response rather than being spent on pinning a parameter. It remains a moment only an endogenous drug price can produce, since a fixed price has no elasticity to match.

3.3 Validation, and what is left undisciplined

None of the moments below is fitted. Each is produced endogenously by a model with sixty single ages, and each is a place the model can be caught.

Moment	Data	Source
Homelessness rate by age, 25 to 34 through 65+	0.316 / 0.338 / 0.291 / 0.250 / 0.077 percent	HUD PIT 2024
Shares of the homeless across those bands	26.0 / 27.2 / 21.0 / 18.4 / 7.5 percent	HUD PIT 2024
Returns to homelessness within 24 months	17.4 percent	HUD SPM 2024
Drug poisonings as a share of all deaths, 25 to 44	about 20 percent	CDC WONDER
Bottom half's share of net worth	2.52 percent	Fed DFA
Housing costs above 30 and above 50 percent of income	40.6 and 17.5 percent	SCF 2022
Age weighted standardised mortality ratio, SUD	2.5 to 9.8	Degenhardt et al.
Age weighted standardised mortality ratio, homeless	9.1 / 4.5 / 1.1 by band	Baggett et al.
Tenant contribution rate	31.2 percent	HUD
Voucher holders with wages as their main income	28 percent	HUD
Income with a moderate or severe SUD, over average	71.1 percent	NSDUH 2024
Labour force participation, SUD against clean	65.8 against 61.5 percent	NSDUH 2024
Homeless at treatment admission	14.5 percent	TEDS-A 2022
Co-occurring psychiatric problem at admission	43.1 percent	TEDS-A 2022
Working at all, among the homeless in treatment	6.6 percent	TEDS-A 2022
Drug use disorder share of the homeless	21.7 percent	Gutwinski et al. (2021)
Drug price participation elasticity	-0.63	Saffer and Chaloupka (1999)

Three parameters remain undisciplined, and they are flagged rather than hidden: the treatment effect on recovery g, the recovery multiplier for SMI μ_m, and the drug supply curvature ν_2. Of these only g carries a headline result, and it is reported as a range rather than as a number.

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Data: SAMHSA (NSDUH 2024, TEDS-A 2022); HUD (Point in Time 2023, Picture of Subsidized Households); BLS (CPS); Census (ACS 2023); Federal Reserve (SCF 2022); CDC (WONDER 2023); SSA; USDA.